

SHEET INDEX

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- GENERAL NOTES (numbered)
1. Design basis: EN 1990 / 1991-1-1 / 1991-1-4 / 1993-1-1 / 1993-1-8 / 1998-1, EN 1992-1-1 8.4 and EN 1992-4 for the anchorage. Site Tripoli, Libya. Design report C Rev 6a, load basis Rev 3, bases_C.md Rev 8, bracing Rev 5a. Roof: one plane at 6 % falling north, TOS(y) = 3.33 + 0.06 (35.87 - y); primary top TOS + 0.03; column top TOS - 0.29; clear height 3.03 m under the cap-plate nuts at C1/C2, 3.07 m under the eave primary.
2. Load basis Rev 3: panel 0.12, purlins 0.05, services 0.10 kN/m2 (permanent); imposed category H q_k 0.40 kN/m2 (psi 0), Q_k 1.0 kN; no wall weight in G (wall mass 0.30 kN/m2 in the seismic mass); G_min 0.17 + steel; gutter 0.25 / -0.50 kN/m; upstands 0.30 kN/m. Wind v_b 27 m/s, q_p by direction: N 1.25, E / W 1.05, S 0.75 kN/m2 (flat-roof coefficients, e = 18 m, z_e 9 m), c_pi +0.2 / -0.3. Seismic a_g 0.10 g, S 1.2, q 1.5, two-mass floor amplification MANDATORY: design roof-level force 133 kN E-W / 134 kN N-S (bays and bases seismic-governed). Temperature +/-30 K service, +45/-25 K erection. Combinations EN 1990 6.10.
3. Materials: hot-rolled S275 J0 (EN 10025-2); shear keys SHS 90x90x8 and Key B bars S355; plates S275; cold-formed Z200x2.0 / C sections S350GD+Z275; PIR sandwich panel 50 mm; BoardX wall panels (0.30 kN/m2, product data to be confirmed); non-shrink cementitious grout C50 class; post-installed bars dia 16 B500 with an EAD 330087 injection system.
4. Bolts: 8.8 (EN 15048 / ISO 4014-4032) hot-dip galvanised, snug tight + 1/4 turn where noted, no preload relied upon; M20 in 22 round holes, M16 in 18, M12 in 14, M24 rods in 26 with turnbuckles and lock nuts; NO slotted holes (thermal path designed for).
5. Welding: shop fillet welds a5-a8 as detailed (E42 consumables, EN 1090-2 EXC2); no site welding on galvanised steel; rod gussets shop-welded to the primaries and bolted to the rafters.
6. Galvanising: hot-dip EN ISO 1461 (85 um) after fabrication for all members, plates and gussets; site touch-up zinc-rich; Z / C sections Z275.
7. Tolerances (EN 1090-2 class 1): base position +/- 10 mm, plumb h/300 (max 10 mm), primary level +/- 5 mm at the cap, rafter TOS +/- 10 mm; bar pattern +/- 15 mm from the scan, never closer than 20 mm to a bar; grout bed 25 +/- 5 mm; column lengths NOMINAL - cut to the surveyed plate-top level.
8. Erection sequence: (1) slab-top survey (datum), scans, cores, proof tests, acceptance per head; (2) core the key pockets, set shims and level; (3) columns C1-C27 and WP1 on the keys, grout pockets and beds, drill and set the bars through the plates, cure, nuts snug + 1/4 turn; (4) primaries P1-P19 on the cap plates (4 M20, tie plates rows F and B), temporary guys; (5) rafters R1-R11, T2, ST1/ST2 on fin plates FP1/FP2; (6) wall bays B1-B7, B9 and the 13 rod panels (turnbuckles snug, lock nuts), release guys; (7) TEMPORARY PLAN BRACING (crossed wire ropes or angles) in one rafter cell of each south band (west wing R2-R3 / y 15.9-21.8, east wing R7-R8 / y 20.1-29.3) or guyed wall columns, kept until the panels and the purlin bridging are complete; (8) purlins, fly braces, eave rails, girts, anti-sag rows; (9) roof panels from the south (high) edge north, well upstands and flashings, gutters and downpipes, wall panels and sills. No panel before all permanent bracing is in.
9. Site verification before any drilling: slab-top level survey; cover-meter scan of all four faces of every column head (top 600 mm: bars, links, cover); GPR from above at all 27 heads (slab solid and 300 mm thick under the plates and pockets, slab top bars); 3-6 cores for the concrete grade; 10 mm pilot drills with feed monitoring; proof tests of 3 bars to >= 60 kN (2 min, <= 1 mm); scan sheet with the final pattern filed per head (S05).
10. Existing structure: the roof adds about 257 kN characteristic gravity (+ 116 kN if the walls are counted), 62 kN characteristic roof-level wind and 133 / 134 kN design seismic force into the 27 pinned 20 x 40 concrete columns and foundations; as-built drawings, the adequacy statement of the existing columns / foundations and the roof survey are client / structural open items to be closed before the permit.
11. Open items: v_b 27 m/s and the terrain sectors to be confirmed (LNMCI); purlin uplift capacity >= 9 kNm and sleeved girt capacities (supplier); BoardX data and drift limit; ambient-vibration period of the existing building (seismic amplification); door-free bays B1-B7, B9 and R6 over the shaft east wall (client); gutter manufacturer data, yard gullies, well outlets and hall-side well walls (architect / mechanical); notch edge beam at WP1 (GPR); client option B1 for the 7 interior columns (bases_C.md appendix).

LEGEND

Primaries IPE 300 (P13 IPE 330)

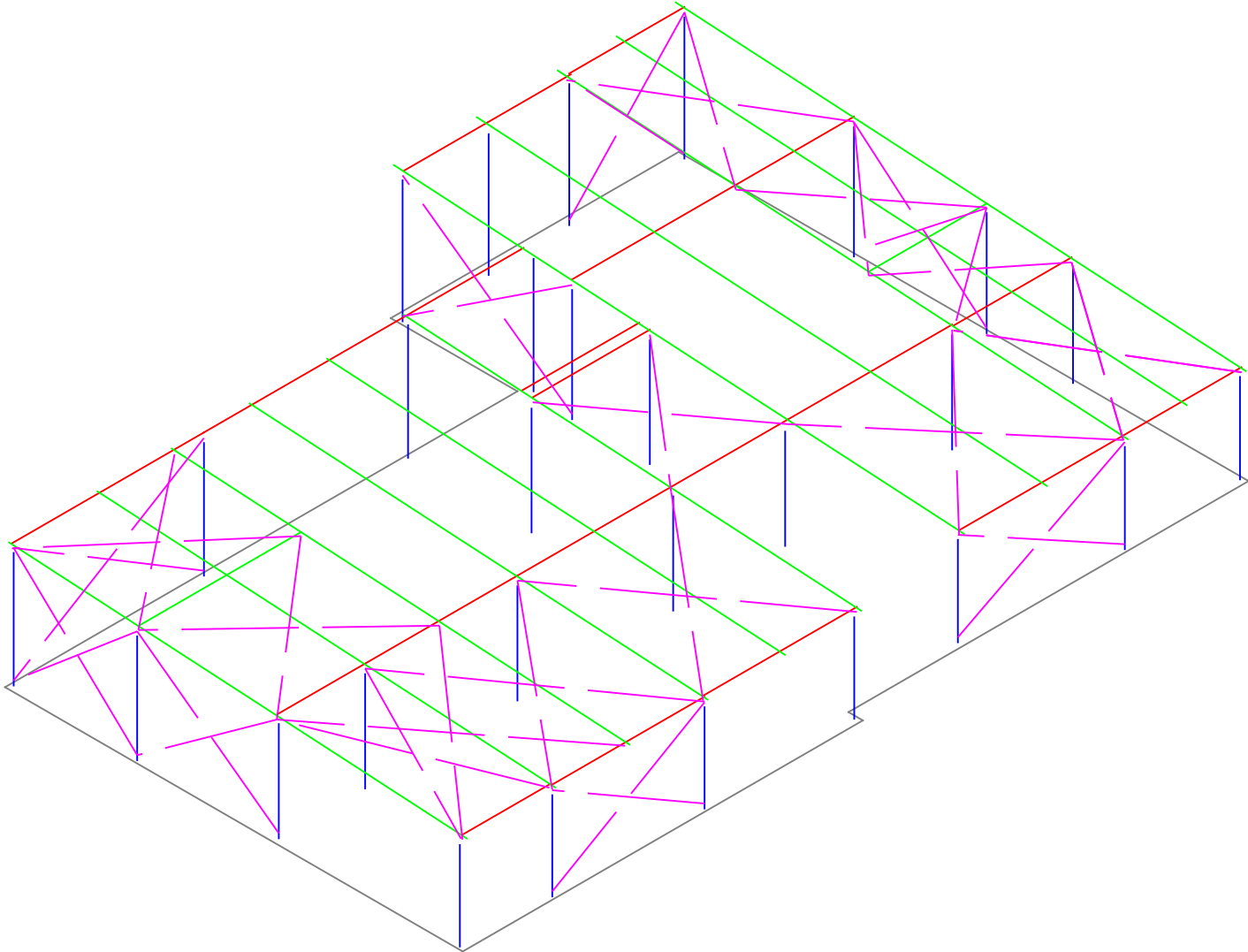
Rafters IPE 240

Columns HEA 140

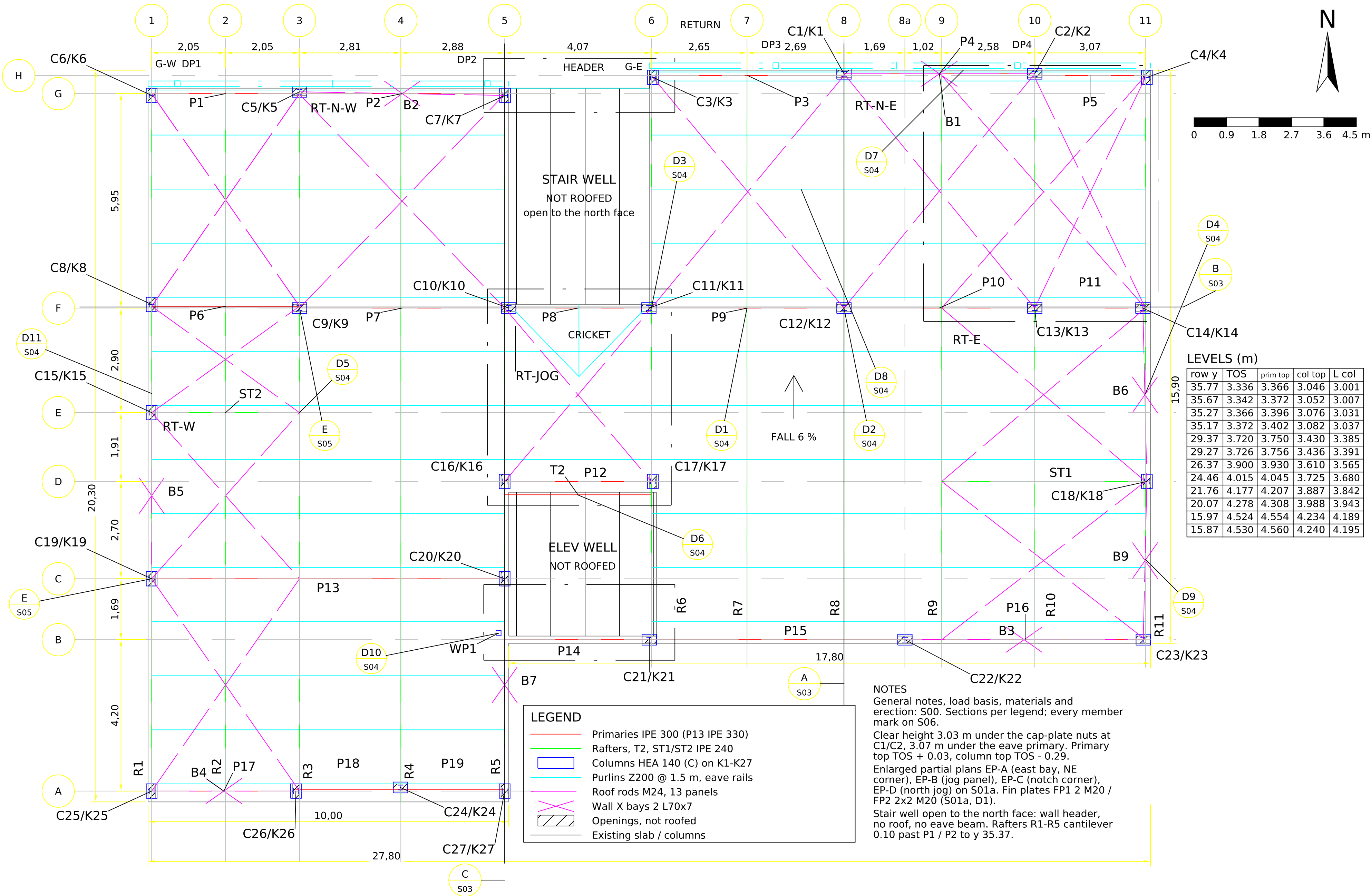
Wall bays / roof rods

Existing slab outline

KEY VIEW - isometric from the NE, panel and purlins hidden; renders in roof/3d/iso_NE_no_panel.png

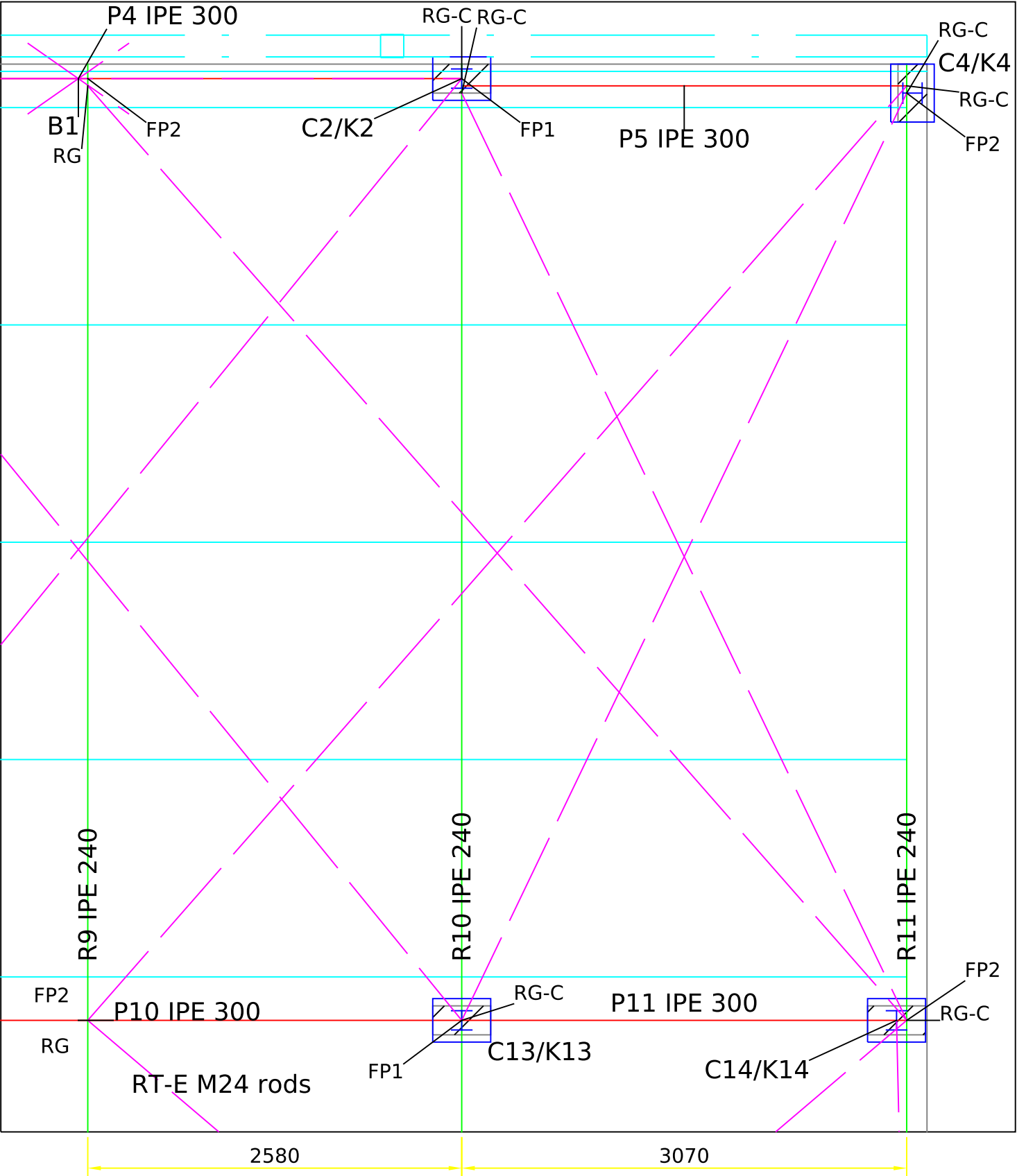


PROJECT Steel roof - Administration Building, Regatta Tourist Village, Tripoli ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m	SHEET TITLE GENERAL NOTES AND SHEET INDEX Not to scale; all sheets 42 x 30 m frames side by side in model space (y 8-38) Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9	REVISIONS			CURRENT REVISION Rev 6a superstructure Rev 8 bases / bracing Rev 5a DATE 2026-09-29 Drawn: Designed: Checked: Approved:	DRAWING No. RTV-ST-S00 S00
		3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a		
		2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)		
		1	2026-09-27	First issue: design Rev 2, bases Rev 3		

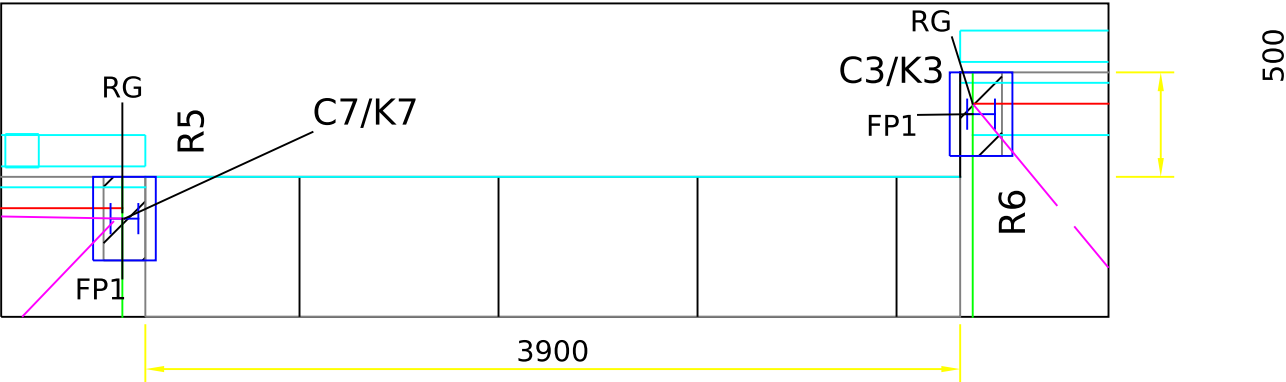


PROJECT Steel roof - Administration Building, Regatta Tourist Village, Tripoli ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m	SHEET TITLE ROOF FRAMING PLAN Scale 1:1 in model space (m); print 1:200 on A1 Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9	REVISIONS			CURRENT REVISION Rev 6a superstructure Rev 8 bases / bracing Rev 5a DATE 2026-09-29 Drawn: Designed: Checked: Approved:	DRAWING No. RTV-ST-S01 S01
		3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a		
		2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)		
		1	2026-09-27	First issue: design Rev 2, bases Rev 3		

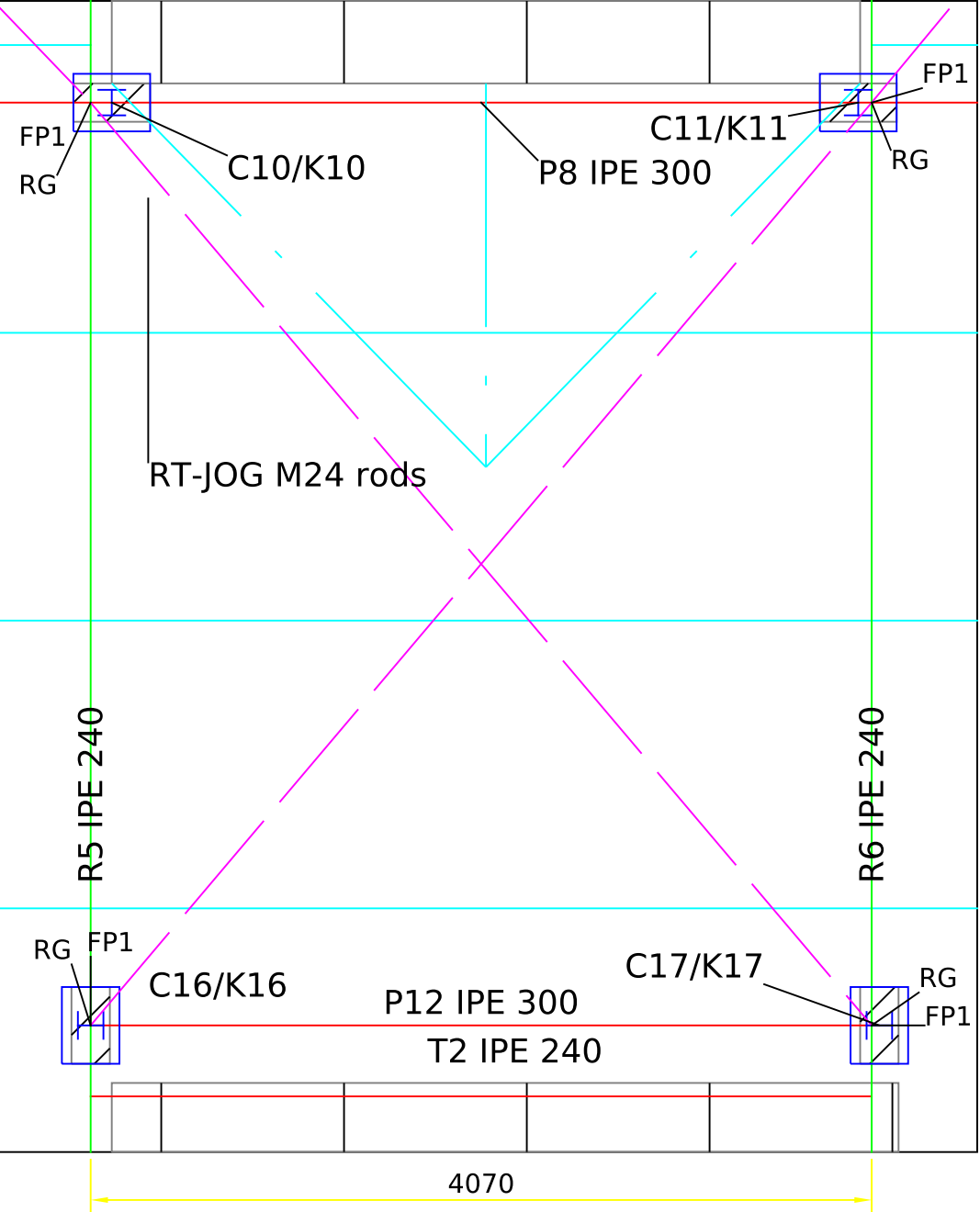
EP-A EAST BAY R9-R11 + NE CORNER



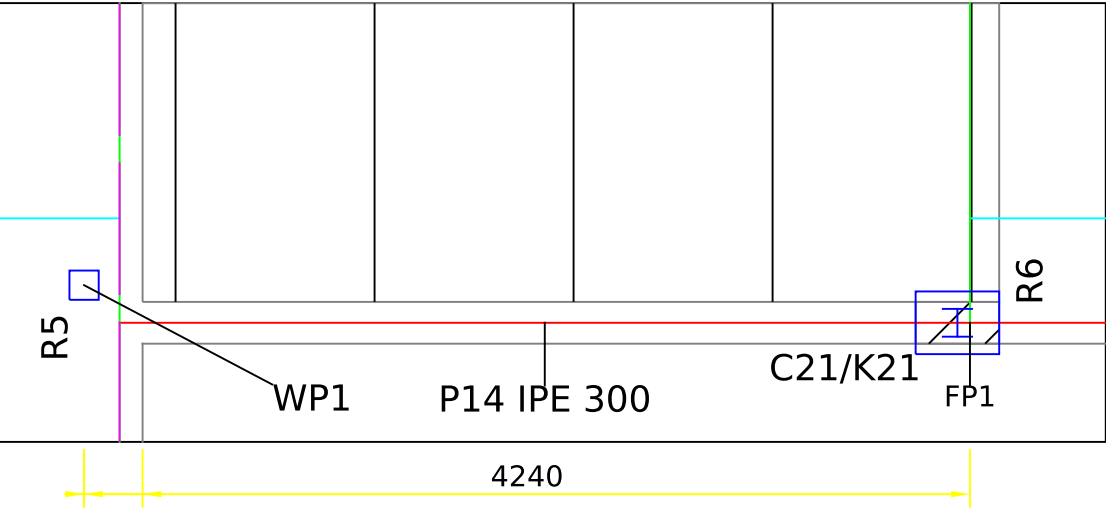
EP-D NORTH JOG, HEADER, RETURN



EP-B JOG PANEL, T2, P12



EP-C NOTCH CORNER, WP1, K21



NOTES TO THE PARTIAL PLANS
FP1: fin plate 100 x 150 x 10, 2 M20 8.8 (pitch 70, e1 = e2 = 40), bolt line 50 from the primary web, welds 2 x a6 (D1). FP2: 160 x 150 x 10, 2 x 2 M20 (p2 60) at the strip-chord splices of R1 / R3 / R9 / R11 (D1). R5 strut fin: FP1 (2 M20, 0.55).
RG: rod gusset 8 mm shop-welded to the primary web, bolted 2 M16 to the rafter web, at every corner of the 13 rod panels (D5). RG-C: combined gusset at C2, C4, C13, C14 where the north strip and the east strip share the NE corner cell (both rod sets, D5).
Rods M24 8.8 with turnbuckles; on RT-W (R1/R3) and RT-E (R9/R11) the rods span two rafter cells and pass R2 / R10 through a bolted web clip (D5).
EP-C: WP1 at (77.61, 20.25); P14 IPE 300 eave beam from R5 (fin plate) to C21; K21 pier 200 thick between the notch edge y 19.97 and the shaft opening y 20.17 (base P, S05).
EP-D: the stair well is open to the north face; wall header 2 x C200x60x2.5 boxed between C7 and the return post; 0.5 m wall return at x 81.79 on brackets from C3; gutter G-W stops at x 77.89, G-E starts at x 81.79 (stop ends + overflow spouts).
Marks: Rn rafters IPE 240, Pn primaries IPE 300 (P13 IPE 330), Cn/Kn columns HEA 140 on concrete column Kn, RT-* rod strips, Bn wall X bays. Sections and lengths on S06.

LEGEND	
—	Primaries IPE 300
—	Rafters IPE 240
	Columns HEA 140
—	Roof rods M24
—	FP1/FP2 fin plates, RG gussets

PROJECT
Steel roof - Administration Building, Regatta Tourist Village, Tripoli
ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m

SHEET TITLE
ENLARGED PARTIAL PLANS EP-A TO EP-D (2x)
Partial plans at 2x (1 m model = 500 mm); dimension text in mm (dimlfac 500)
Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9

REVISIONS			
3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a	
2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)	
1	2026-09-27	First issue: design Rev 2, bases Rev 3	

CURRENT REVISION
Rev 6a superstructure
Rev 8 bases / bracing Rev 5a
DATE 2026-09-29
Drawn: Designed:
Checked: Approved:

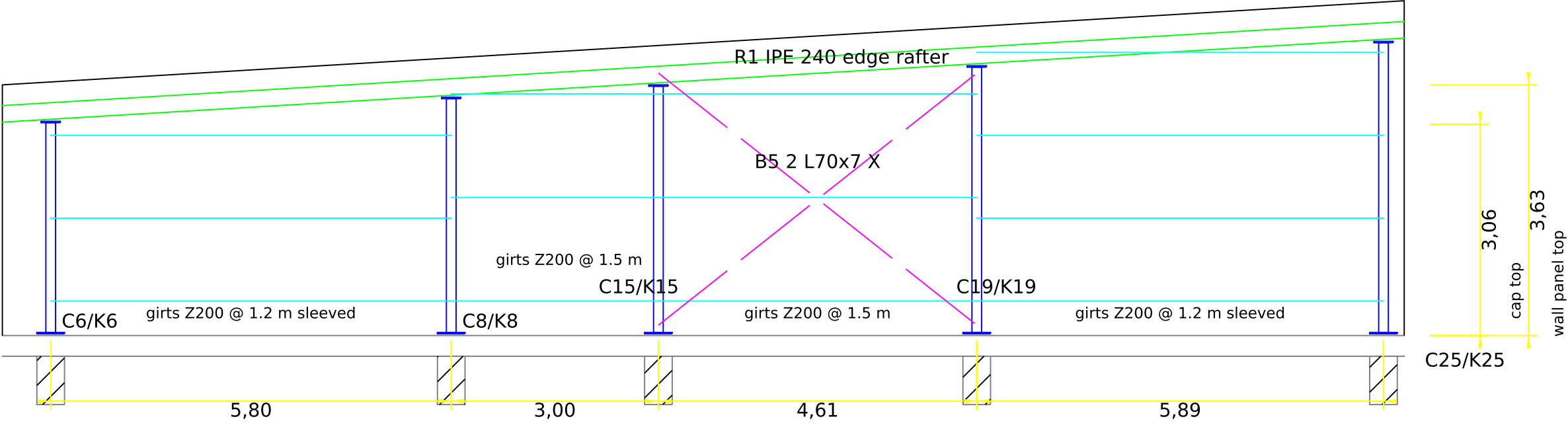
DRAWING No.
RTV-ST-S01a
S01a

COLUMN SCHEDULE (HEA 140 S275)

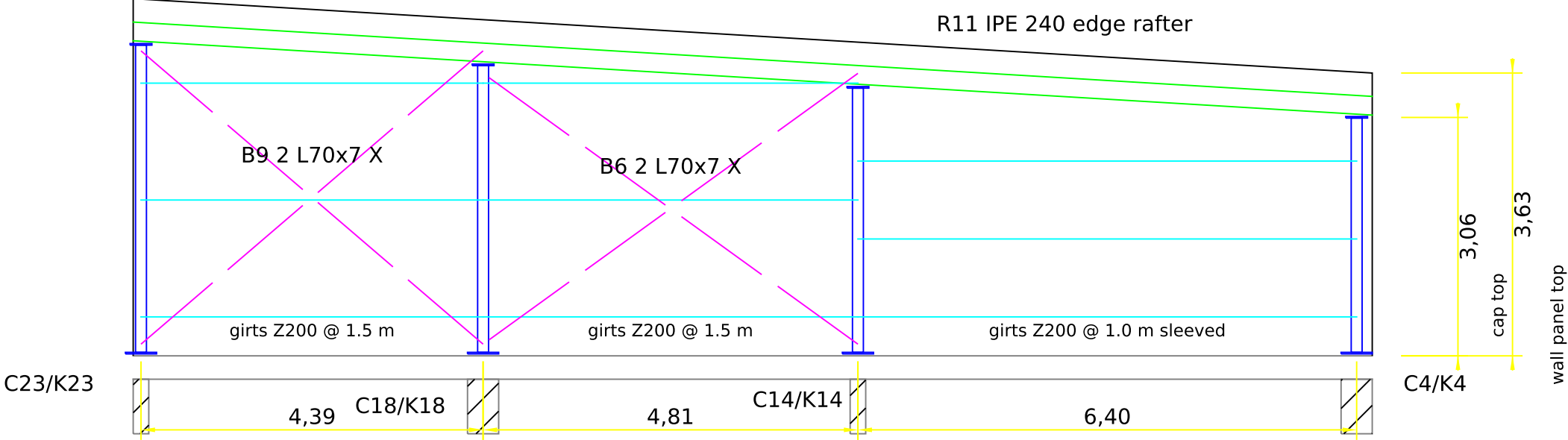
Mark	Conc.	Section	x (m)	y (m)	L col	Col t.	Plate top	Ba.	Conc. col.	HEA web	Bays	Face
C1	K1	HEA 140	87.19	35.77	3.001	3.046	+0.045	E	E-W 400x2.	N-S	B1	N2
C2	K2	HEA 140	92.48	35.77	3.001	3.046	+0.045	E	E-W 400x2.	N-S	B1	N2
C3	K3	HEA 140	81.89	35.67	3.007	3.052	+0.045	E	N-S 200x4.	E-W	-	N2
C4	K4	HEA 140	95.59	35.67	3.007	3.052	+0.045	E	N-S 200x4.	E-W	-	N2,E
C5	K5	HEA 140	72.09	35.27	3.031	3.076	+0.045	E	E-W 400x2.	N-S	B2	N1
C6	K6	HEA 140	67.99	35.17	3.037	3.082	+0.045	E	N-S 200x4.	E-W	-	N1,W
C7	K7	HEA 140	77.79	35.17	3.037	3.082	+0.045	E	N-S 200x4.	E-W	B2	N1
C8	K8	HEA 140	67.99	29.37	3.385	3.430	+0.045	E	N-S 200x4.	E-W	-	W
C9	K9	HEA 140	72.09	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	-	int
C10	K10	HEA 140	77.89	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	-	int
C11	K11	HEA 140	81.78	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	-	int
C12	K12	HEA 140	87.19	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	-	int
C13	K13	HEA 140	92.48	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	-	int
C14	K14	HEA 140	95.48	29.27	3.391	3.436	+0.045	E	E-W 400x2.	N-S	B6	E
C15	K15	HEA 140	67.99	26.37	3.565	3.610	+0.045	E	N-S 200x4.	E-W	B5	W
C16	K16	HEA 140	77.78	24.46	3.680	3.725	+0.045	E	N-S 200x4.	E-W	-	int
C17	K17	HEA 140	81.89	24.46	3.680	3.725	+0.045	E	N-S 200x4.	E-W	-	int
C18	K18	HEA 140	95.59	24.46	3.680	3.725	+0.045	E	N-S 200x4.	E-W	B6,B9	E
C19	K19	HEA 140	67.99	21.76	3.842	3.887	+0.045	E	N-S 200x4.	E-W	B5	W
C20	K20	HEA 140	77.78	21.76	3.842	3.887	+0.045	E	N-S 200x4.	E-W	B7	int
C21	K21	HEA 140	81.79	20.07	3.943	3.988	+0.045	P	E-W 400x2.	N-S	-	S2
C22	K22	HEA 140	88.88	20.07	3.943	3.988	+0.045	E	E-W 400x2.	N-S	B3	S2
C23	K23	HEA 140	95.49	20.07	3.943	3.988	+0.045	E	E-W 400x2.	N-S	B3,B9	S2,E
C24	K24	HEA 140	74.89	15.97	4.189	4.234	+0.045	E	E-W 400x2.	N-S	-	S1
C25	K25	HEA 140	67.99	15.87	4.195	4.240	+0.045	E	N-S 200x4.	E-W	B4	S1,W
C26	K26	HEA 140	71.99	15.87	4.195	4.240	+0.045	E	N-S 200x4.	E-W	B4	S1
C27	K27	HEA 140	77.78	15.87	4.195	4.240	+0.045	E	N-S 200x4.	E-W	B7	S1,EN
WP1	-	HEA 140	77.61	20.25	4.24	slott.	+0.040	post	edge beam	N-S	-	S2,EN

SCHEDULE NOTES
Column top = TOS - 0.29 (cap-plate underside); L nominal = TOS - 0.335 (25 grout + 20 plate). Column lengths are NOMINAL: cut columns to the surveyed plate-top level from the slab-top survey (S00 section 9); grout bed 25 +/- 5 mm.
Clear height 3.03 m under the cap-plate nuts at C1/C2, 3.07 m under the eave primary. Bases type E (P at K21) per S05; base plate top +0.045.
Girts Z200x2.0 on cleats (D9); rows per bay as noted; bottom row 0.50 above the slab; wall sill / base rail D11. Bays B1-B7, B9 door-free (client).

W WEST ELEV. x 67.89, north left

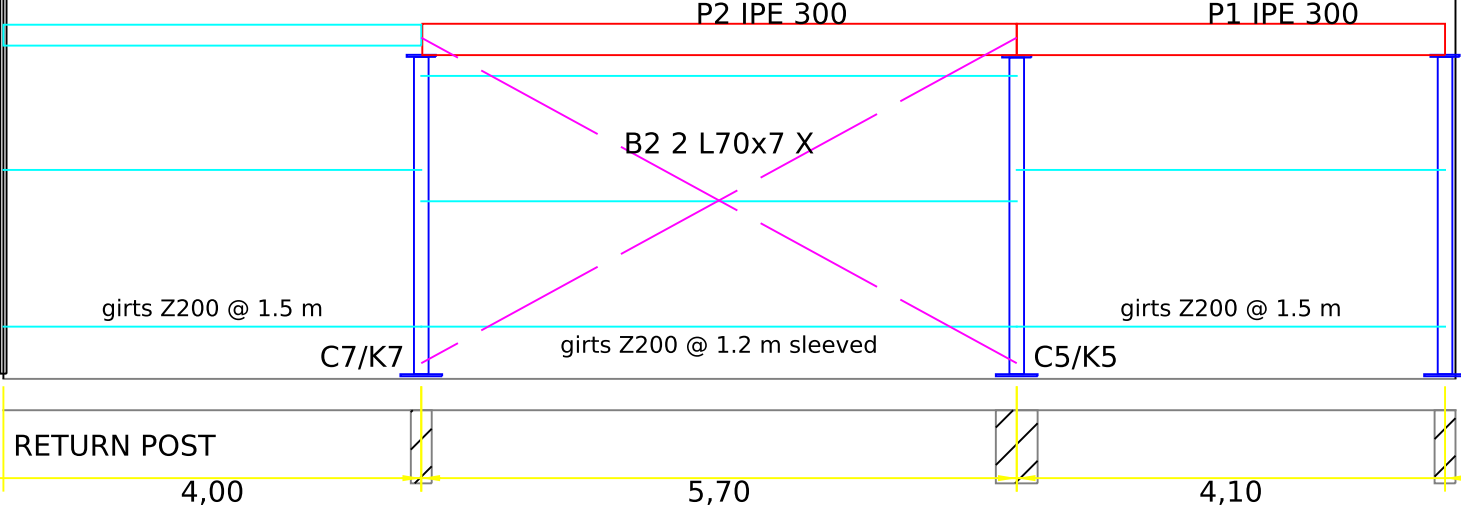


E EAST ELEV. x 95.69, north right

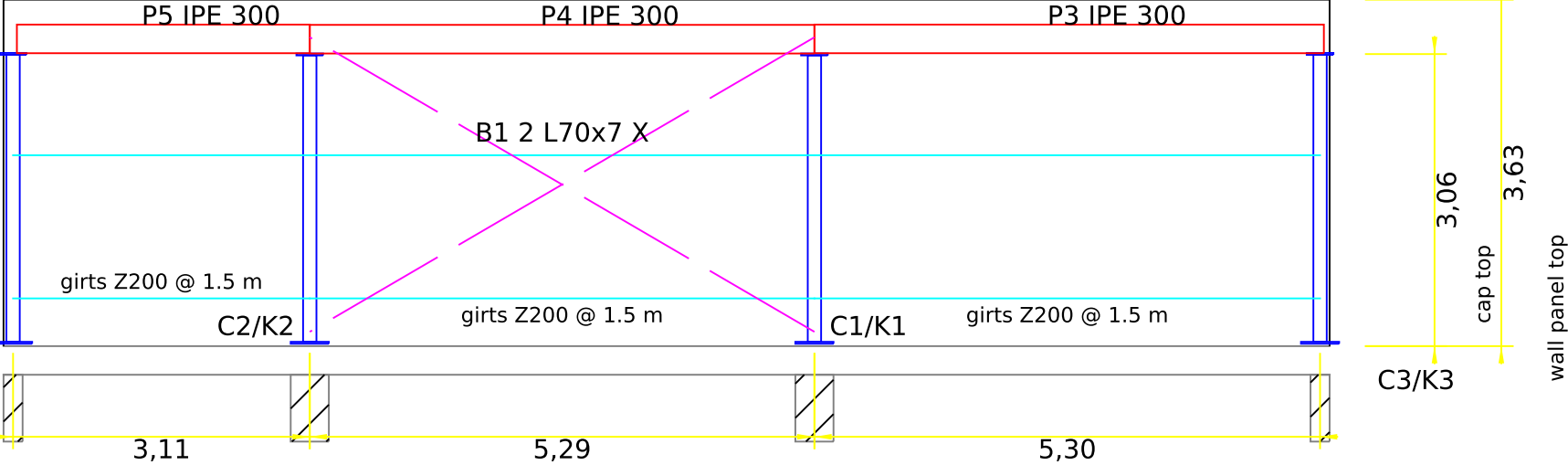


WALL HEADER 2 C200x60x2.5

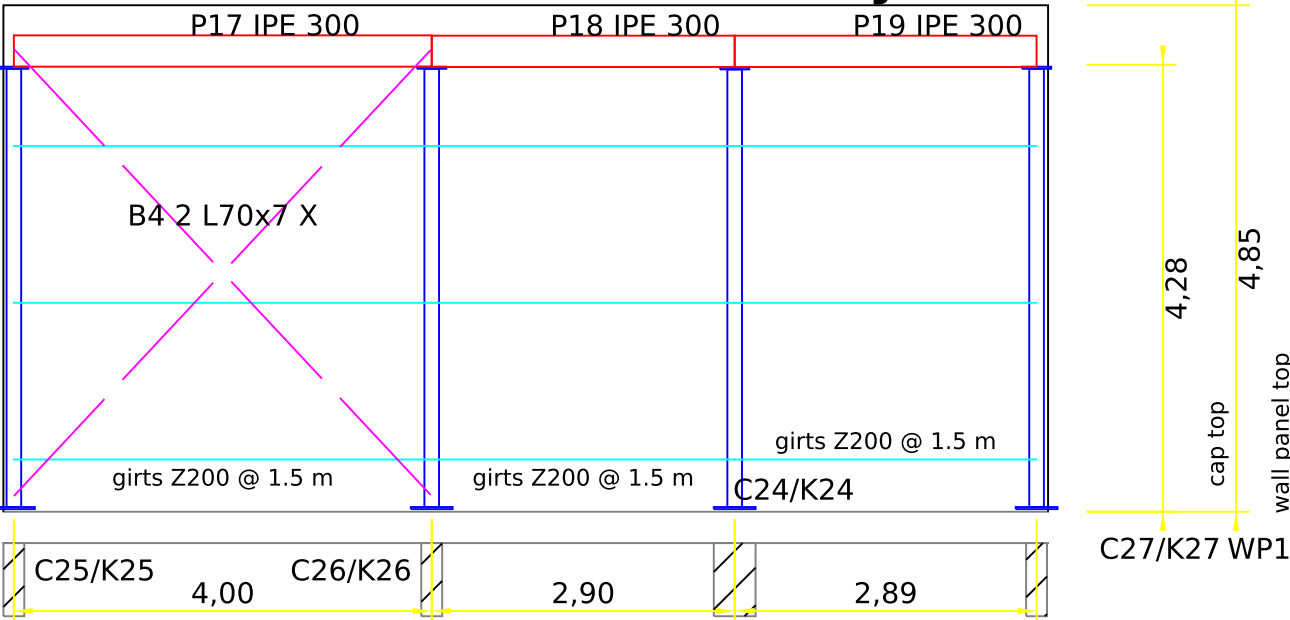
N1 NORTH ELEV. WEST y 35.37



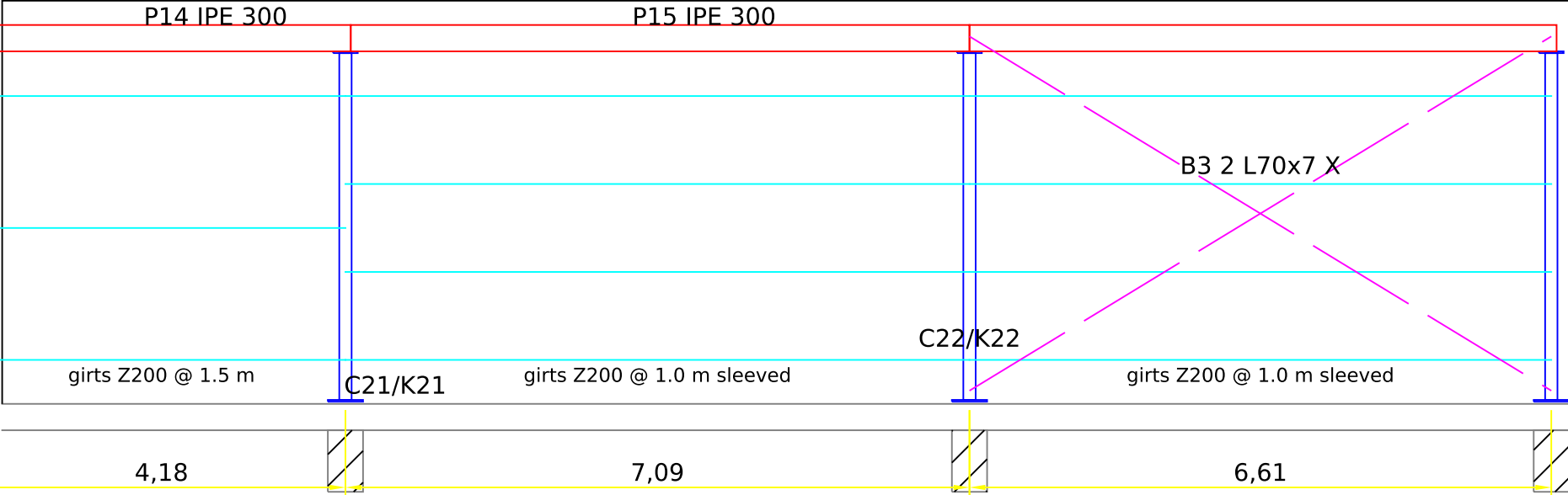
N2 NORTH ELEV. EAST y 35.87



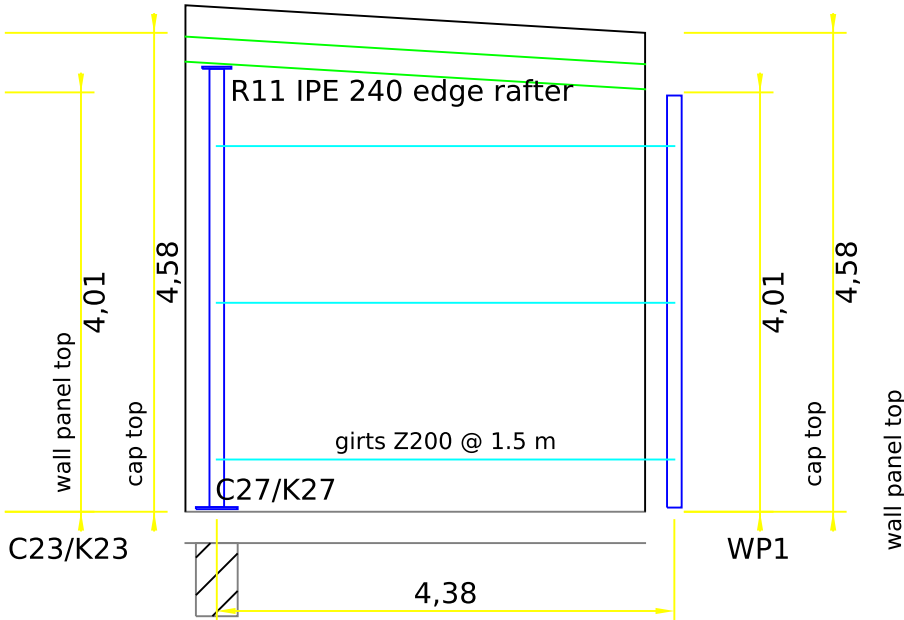
S1 SOUTH ELEV. WEST y 15.57



S2 SOUTH ELEV. EAST y 19.97



EN NOTCH FACE



PROJECT
Steel roof - Administration Building, Regatta Tourist Village, Tripoli

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Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m

SHEET TITLE
COLUMN SCHEDULE AND WALL ELEVATIONS

Elevations 1:1 in model space (m); print 1:200 on A1
Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9

REVISIONS			
3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a	
2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)	
1	2026-09-27	First issue: design Rev 2, bases Rev 3	

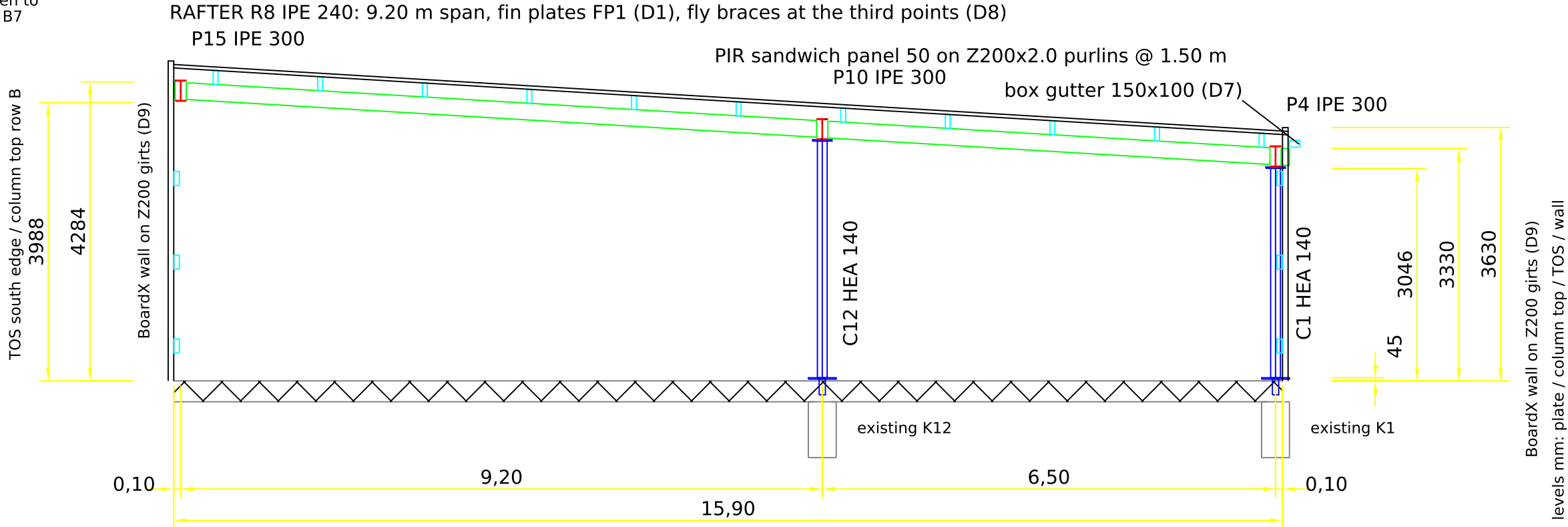
CURRENT REVISION
Rev 6a superstructure
Rev 8 bases / bracing Rev 5a
DATE 2026-09-29
Drawn: Designed:
Checked: Approved:

DRAWING No.
RTV-ST-S02

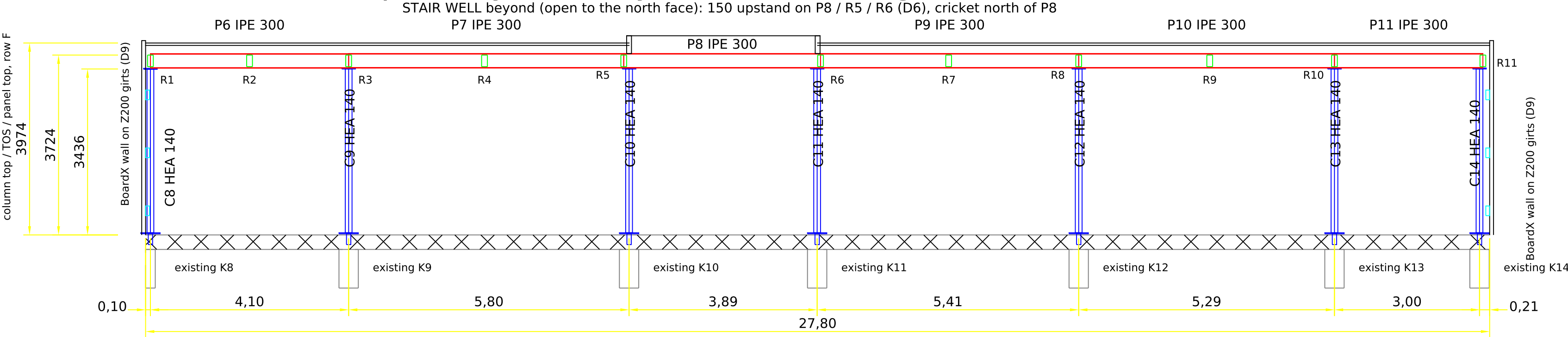
S02

SECTION NOTES
A-A: north on the right; rafter R8 9.20 m span; C1 and C12 in section; gutter G-E at the north eave. Clear height 3.03 m under the cap-plate nuts at C1/C2, 3.07 m under the eave primary P4.
B-B: east on the right; rafters R1-R11 beyond shown dashed; stair well beyond open to the north face. C-C: north on the right; R5 ends at y 35.37 (west north edge); bay B7 C27-C20; WP1 beyond.

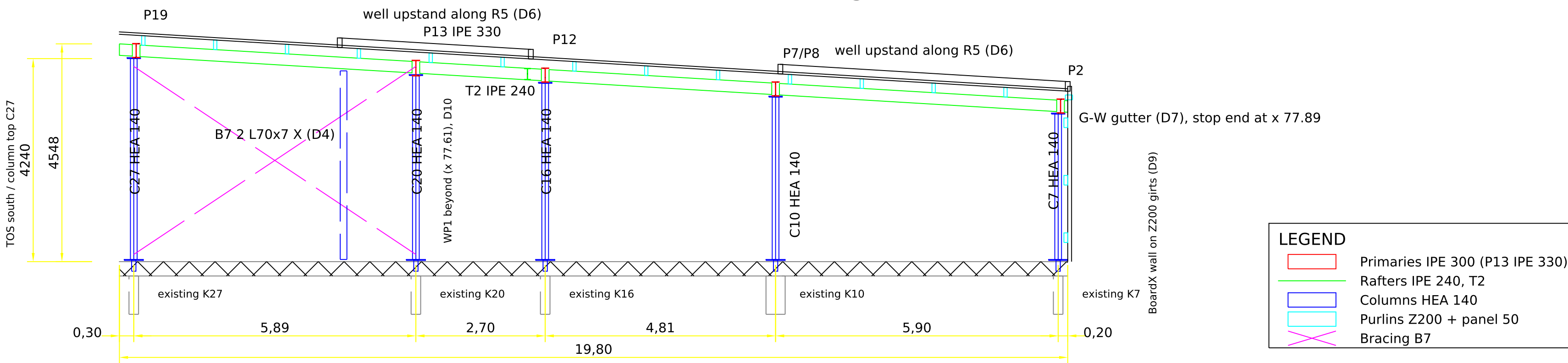
SECTION A-A rafter line 8 (x 87.19), looking west



SECTION B-B primary row F (y 29.27), looking north



SECTION C-C braced line 5 (x 77.78), looking west



- LEGEND
- Primaries IPE 300 (P13 IPE 330)
 - Rafters IPE 240, T2
 - Columns HEA 140
 - Purlins Z200 + panel 50
 - Bracing B7

PROJECT
Steel roof - Administration Building, Regatta Tourist Village, Tripoli

ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m

SHEET TITLE
TYPICAL SECTIONS A-A, B-B, C-C

Sections 1:1 in model space (m); levels in mm; print 1:100 on A1
Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9

REVISIONS			
3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a	
2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)	
1	2026-09-27	First issue: design Rev 2, bases Rev 3	

CURRENT REVISION
Rev 6a superstructure
Rev 8 bases / bracing Rev 5a
DATE 2026-09-29
Drawn: Designed:
Checked: Approved:

DRAWING No.
RTV-ST-S03
S03

PLAN at the primary web: FP2 two-row plate

SECTION along the primary, top TOS + 30

rafter bottom +30

IPE 300 (level)

FP1 100x150x10

rafter (north piece)

primary web / flanges

IPE 240, plumb cut

rafter (south piece)

M20 8.8 in 22 round holes, snug tight, PLAIN plates (no slots); FP1 bolts e 40 / p 70. FP1 at every rafter / primary junction; FP2 (2 x 2 M20, p2 60) where the rafter is a strip chord: R1, R3, R9, R11; R5 strut splice FP1 (0.55). Web bearing on the 6.2 mm IPE 240 web governs: 64.6 kN per M20 (one row), 54.8 kN (two rows); max gravity reaction 21.3 kN (0.29). Force passes plate - weld - primary web - weld - plate.

D1

S04

FIN PLATES

scale 4x

FP1 100x150x10 2 M20 8.8; FP2 160x150x10 2x2 M20 (p2 60) at the strip-chord splices R1/R3/R9/R11; IPE 240 to IPE 300 web

ELEVATION normal to the primary (end column)

IPE 300, level, no pack

cap top = TOS - 270

HEA 140 column (flange 140)

PLAN of the cap plate

HEA 140 below

Cap plate 200 x 280 x 20 S275, holes 22; bolt rows 33 mm outside the 133 column depth (cantilever 0.11), Tension 44.6 kN (K12 uplift); bolt 0.08, flange T-stub 0.12; chord force 70.2 kN; interaction 0.24. Primaries bolted before the rafters are landed. No stiffeners. Slope is in the rafters only; primaries and cap plates are level.

D2

S04

CAP PLATE

scale 4x

HEA 140 cap plate 200x280x20, a6 all round; 4 M20 8.8 through the IPE 300 bottom flange, gauge 90, pitch 200 (nuts clear the rafter flange)

ELEVATION along the primary row

primary end

primary end

tie plate 140x400x10

gap 20

rafter

PLAN: rafter continuity through the primary web

cap plate below

rafter

Chord force <= 70 kN passes through the tie plate and the 4 M20 (2 per primary end); rafter strut force through the fin plates and the primary web. Rows A / G / H (eave beams): no tie plate.

D3

S04

CHORD TIE

scale 4x

Primary continuity over an interior column (rows F and B): 10 mm tie plate 140 x 400 under both primary ends; gap 20

ELEVATION in the wall plane, column base

HEA 140 flange 140

a6

gusset 10 mm

L70x7 diagonal

e1 40, p1 110, e2 30

grout 25

column CL

Gusset 10 mm S275 in the wall plane; angle net section 189 kN, 2 M20 shear 188 kN, gusset bearing 2 x 124 kN; max T_Ed 76.1 kN (B7) -> 0.41. Diagonal centrelines meet at the column CL at base-plate top and at the primary centre. X in every bay: both diagonals, crossing clipped with a 10 mm plate and 1 M16. Eave gusset: same plate welded to the column flange and the cap-plate underside.

D4

S04

BRACE GUSSET

scale 4x

L70x7 tension-only diagonals, 10 mm gusset welded a6 to the column flange and the base plate (eave: cap-plate underside); 2 M20 8.8 per angle end

PLAN at a panel corner

rafter IPE 240

turnbuckle M24

web hole 60 + ring 8

hole 26, nut + lock nut

gusset 8, a5 to the primary

primary IPE 300

leg 2 M16 to the rafter web

Rods M24 8.8 (F_t,Rd 203 kN), max T 80.8 kN (RT-JOG, 0.40); gussets only at the 13 panel corners (RG). RG-C at C2, C4, C13, C14: one 10 mm gusset carries both rod sets (RT-N-E + RT-E, 59.6 + 0.3 x 53.1 kN), 2 M20 per rod end. No site welding on galvanised steel. RT-W / RT-E rods span two rafter cells: R2 / R10 passed through a bolted web clip; sag ties to the purlins at the crossing and at 3 m.

D5

S04

ROOF ROD GUSSET

scale 4x

M24 8.8 rods with turnbuckles; 8 mm gusset SHOP-welded a5 to the primary web, BOLTED 2 M16 to the rafter web; combined gusset RG-C at the NE corner

SECTION across trimmer T2, looking east

upstand C100x50x3

cap flashing 0.7

2 M12 to the seat

seat plate 15

PIR panel 50

edge purlin Z200

100 panel stop / closer

trimmer T2 IPE 240

WELL (open)

ROOF

Upstand >= 150 above the panel top on all sides of both wells; soakers at panel seams; well-side face lined with the same coated sheet. Stair well: open to the north face (no trimmer); cricket on P8 (ridge 120, upstand 270-300).

D6

S04

WELL UPSTAND

scale 4x

T2 IPE 240 trimmer on FP1 to R5/R6; C100x50x3 upstand 150 above the panel; cap flashing and panel stop; same on R5/R6 and P8 (cricket side 270-300)

SECTION at the north eave, north on the right

purlin Z200 @ 300

PIR panel 50

rafter ends +100

top girt Z200

fascia flashing

eave rail C200x60x2.5, 2 M12

bracket 40x5 @ 600

HEA 140 (D2)

BoardX wall (D9)

box gutter 150x100

drip flashing 0.7, lap 60

Gutter 0.7 galvanised + polyester (or 1.0 alu), riveted + butyl joints, fall 1:350 to DP1-DP4, EPDM expansion joint at each high point; stop ends and overflow spouts 100x30 at x 67.89 / 77.89 / 81.79 / 95.69. Loads 0.25 kN/m gravity, -0.50 kN/m uplift (zone F), 0.5 kN point at any bracket; fixed brackets at the outlets, sliding elsewhere. Rafters R1-R5 cantilever 0.10 past P1/P2 to y 35.37.

D7

S04

EAVE GUTTER

scale 4x

Rafter end, eave rail C200x60x2.5, brackets 40x5 @ 600, box gutter 150x100, fascia, first purlin, panel; two runs G-W (y 35.37) / G-E (y 35.87)

SECTION across the rafter (looking north)

PIR panel 50

fly brace L50x5

Z200x2.0 purlin

cleat 120x160x8, a4

rafter IPE 240

Purlins simple span between rafters @ 1.5 m, one mid-span anti-sag row per span; fly braces at mid-span (L <= 6.6 m) and third points (7.5 / 9.2 m). Cleat uplift ~11 kN in zone F: 2 M12 8.8 (0.35). Supplier uplift capacity >= 9 kNm before order.

D8

S04

PURLIN CLEAT

scale 5x

Z200x2.0 on a plate cleat 120x160x8, 2 M12; fly brace L50x5 from the rafter bottom flange to the purlin; anti-sag row at mid-span

PROJECT

Steel roof - Administration Building, Regatta Tourist Village, Tripoli

ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS

Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m

SHEET TITLE

CONNECTION DETAILS D1 - D8

Details drawn 4x or 5x in model space (scale in each frame); DIMENSION text = true mm (dimlfac 250 / 200)

Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9

REVISIONS

3	2026-09-29	Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a
2	2026-09-28	Design Rev 3 / bases Rev 4b (critique C1-C15)
1	2026-09-27	First issue: design Rev 2, bases Rev 3

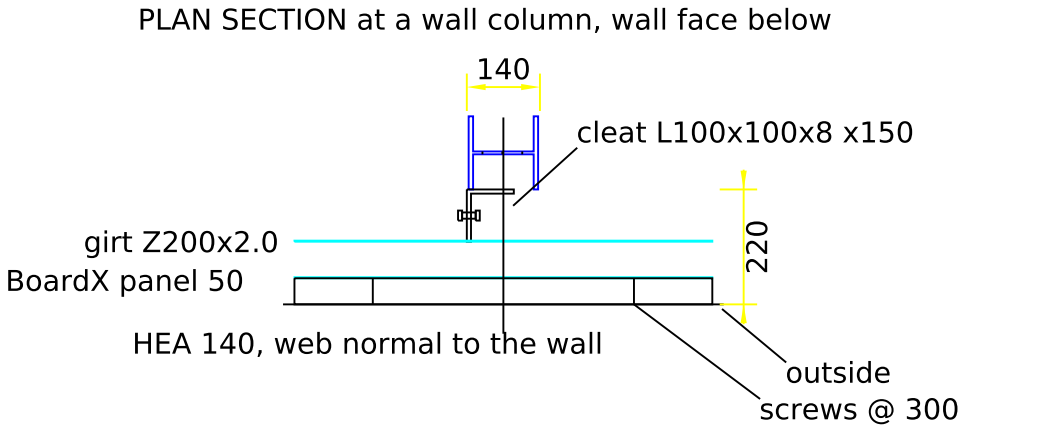
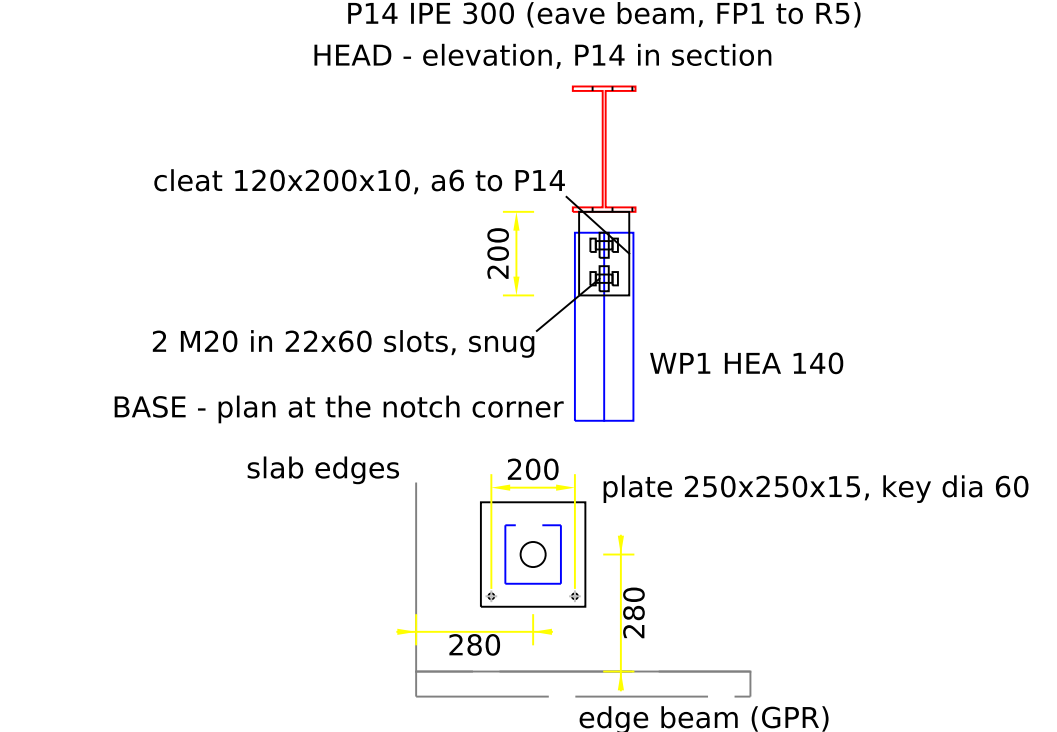
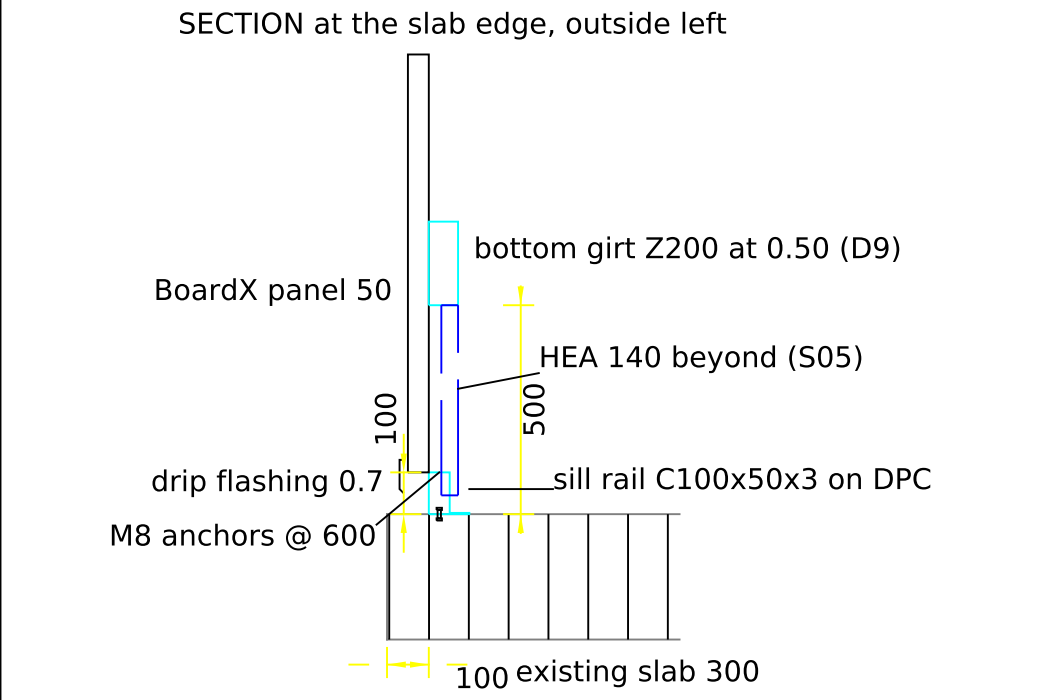
CURRENT REVISION

Rev 6a superstructure
Rev 8 bases / bracing Rev 5a
DATE 2026-09-29
Drawn: Designed:
Checked: Approved:

DRAWING No.

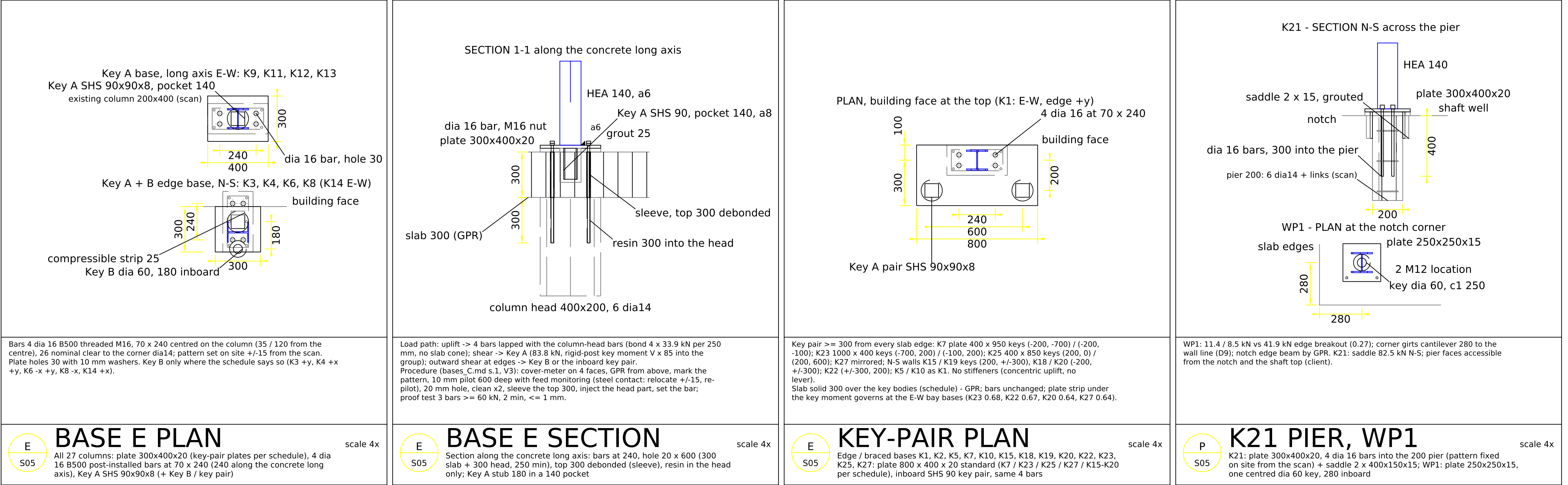
RTV-ST-S04

S04

PLAN SECTION at a wall column, wall face below 	P14 IPE 300 (eave beam, FP1 to R5) HEAD - elevation, P14 in section 	SECTION at the slab edge, outside left 	Steel S275 J0; bolts 8.8 hot-dip galvanised, snug tight (no preload relied upon); M20 in 22 holes, M16 in 18, M12 in 14, M24 rods in 26; all holes round (no slots). Shop fillet welds a5-a8 as noted (E42 / S275 consumables). NO site welding on galvanised steel: rod gussets shop-welded to the primaries and bolted to the rafters (D5). Thermal: +/-30 K service, +45/-25 K erection; the E-W path B2 - RT-N-W - row F chord - RT-N-E - B1 carries 15 kN (wind) / 37 kN (erection) locked-in; designed for, no slotted holes. Rafter ends plumb-cut, no copes; primaries level on level cap plates (primary top TOS + 30, rafter top TOS); slope 6 % in the rafters only. Precamber 10-15 mm on the 9.2 m rafters R7-R10 (south spans) per design report Rev 6a; none on P13 (IPE 330). Fly braces at mid-span (L <= 6.6 m) and third points (7.5 / 9.2 m); anti-sag row at mid-span of every purlin span; purlin cleats 2 M12 8.8 (uplift ~11 kN, zone F). Erection: temporary plan bracing (crossed wire ropes or angles) in one rafter cell of each south band until the panels and purlin bridging are complete (S00 section 8).
D9 S04 GIRT, PANEL Z200x2.0 girt on an L100x100x8 x 150 cleat to the column flange (2 M12); BoardX panel on the girt outer flange, screws @ 300 scale 5x	D10 S04 WIND POST HEA 140 at (77.61, 20.25), 280 inboard of the notch corner: slotted head to eave beam P14 (2 M20 in 22x60 slots); base 250x250x15, one dia 60 key, 2 M12 scale 4x	D11 S04 WALL SILL BoardX wall base at the slab edge: C100x50x3 sill rail on the slab, M8 anchors @ 600, drip flashing; bottom girt 0.50 above; well walls hall-side: blockwork (architect) scale 4x	GN S04 NOTES General notes for D1-D11; see S00 for materials, bolts, welding, galvanising and tolerances scale 5x

DETAIL INDEX
D1 fin plates FP1 / FP2 (S04) - D2 cap plate - D3 chord tie - D4 bracing gusset - D5 roof rod gusset, RG-C combined gusset, R2 / R10 web clip - D6 well upstand (T2, R5 / R6, P8) - D7 eave and gutter - D8 purlin cleat and fly brace (S04); D9 girt cleat and wall panel - D10 wind post WP1 head and base - D11 wall sill / base rail (this sheet). Base details type E / P / WP1: S05.

PROJECT		SHEET TITLE		REVISIONS		CURRENT REVISION		DRAWING No.	
Steel roof - Administration Building, Regatta Tourist Village, Tripoli		CONNECTION DETAILS D9 - D11, NOTES		3		Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a		Rev 6a superstructure	
				2		Design Rev 3 / bases Rev 4b (critique C1-C15)		Rev 8 bases / bracing Rev 5a	
				1		First issue: design Rev 2, bases Rev 3		DATE 2026-09-29	
ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS		Details drawn 4x or 5x in model space (scale in each frame); DIMENSION text = true mm (dimlfac 250 / 200)				Drawn: Designed:		S04a	
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m		Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9				Checked: Approved:			



MEMBERS (1-34 of 102)

Mark	Section	L m	n	kg	between
T2	IPE 240	4.07	1	125	R78-R82
P14	IPE 300	4.07	1	172	R78-K21
R1.1	IPE 240	5.89	1	181	K25-K19
R1.2	IPE 240	4.61	1	142	K19-K15
R1.3	IPE 240	3.00	1	92	K15-K8
R1.4	IPE 240	5.80	1	178	K8-K6
R2.1	IPE 240	5.89	1	181	P_K25K26-P_K19K20
R2.2	IPE 240	7.56	1	232	P_K19K20-P_K8K9
R2.3	IPE 240	5.90	1	181	P_K8K9-P_K6K5
R3.1	IPE 240	5.89	1	181	K26-P_K19K20
R3.2	IPE 240	7.51	1	231	P_K19K20-K9
R3.3	IPE 240	6.00	1	184	K9-K5
R4.1	IPE 240	5.79	1	178	K24-P_K19K20
R4.2	IPE 240	7.51	1	231	P_K19K20-P_K9K10
R4.3	IPE 240	5.93	1	182	P_K9K10-P_K5K7
R5.1	IPE 240	5.89	1	181	K27-K20
R5.2	IPE 240	2.70	1	83	K20-K16
R5.3	IPE 240	4.81	1	148	K16-K10
R5.4	IPE 240	5.90	1	181	K10-K7
R6.1	IPE 240	4.39	1	135	K21-K17
R6.2	IPE 240	4.81	1	148	K17-K11
R6.3	IPE 240	6.40	1	196	K11-K3
R7.1	IPE 240	9.20	1	282	P_K21K22-P_K11K12
R7.2	IPE 240	6.40	1	196	P_K11K12-P_K3K1
R8.1	IPE 240	9.20	1	282	P_K21K22-K12
R8.2	IPE 240	6.50	1	200	K12-K1
R9.1	IPE 240	9.20	1	282	P_K22K23-P_K12K13
R9.2	IPE 240	6.50	1	200	P_K12K13-P_K1K2
R10.1	IPE 240	9.20	1	282	P_K22K23-K13
R10.2	IPE 240	6.50	1	200	K13-K2
R11.1	IPE 240	4.39	1	135	K23-K18
R11.2	IPE 240	4.81	1	148	K18-K14
R11.3	IPE 240	6.40	1	196	K14-K4
P1	IPE 300	4.10	1	173	K6-K5

MEMBERS (35-68 of 102)

Mark	Section	L m	n	kg	between
P2	IPE 300	5.69	1	240	K5-K7
P3	IPE 300	5.34	1	225	K3-K1
P4	IPE 300	5.29	1	223	K1-K2
P5	IPE 300	3.07	1	130	K2-K4
P6	IPE 300	4.10	1	173	K8-K9
P7	IPE 300	5.69	1	240	K9-K10
P8	IPE 300	4.07	1	172	K10-K11
P9	IPE 300	5.34	1	225	K11-K12
P10	IPE 300	5.29	1	223	K12-K13
P11	IPE 300	3.07	1	130	K13-K14
P12	IPE 300	4.07	1	172	K16-K17
P13	IPE 330	9.79	1	481	K19-K20
P15	IPE 300	7.03	1	297	K21-K22
P16	IPE 300	6.67	1	281	K22-K23
P17	IPE 300	4.00	1	169	K25-K26
P18	IPE 300	2.90	1	122	K26-K24
P19	IPE 300	2.89	1	122	K24-K27
C1	HEA 140	3.00	1	74	K1
C2	HEA 140	3.00	1	74	K2
C3	HEA 140	3.01	1	74	K3
C4	HEA 140	3.01	1	74	K4
C5	HEA 140	3.03	1	75	K5
C6	HEA 140	3.04	1	75	K6
C7	HEA 140	3.04	1	75	K7
C8	HEA 140	3.38	1	83	K8
C9	HEA 140	3.39	1	84	K9
C10	HEA 140	3.39	1	84	K10
C11	HEA 140	3.39	1	84	K11
C12	HEA 140	3.39	1	84	K12
C13	HEA 140	3.39	1	84	K13
C14	HEA 140	3.39	1	84	K14
C15	HEA 140	3.56	1	88	K15
C16	HEA 140	3.68	1	91	K16
C17	HEA 140	3.68	1	91	K17

MEMBERS (69-102 of 102)

Mark	Section	L m	n	kg	between
C18	HEA 140	3.68	1	91	K18
C19	HEA 140	3.81	1	94	K19
C20	HEA 140	3.81	1	94	K20
C21	HEA 140	3.94	1	97	K21
C22	HEA 140	3.94	1	97	K22
C23	HEA 140	3.94	1	97	K23
C24	HEA 140	4.19	1	103	K24
C25	HEA 140	4.20	1	104	K25
C26	HEA 140	4.20	1	104	K26
C27	HEA 140	4.20	1	104	K27
WP1	HEA 140	4.24	1	105	notch corner
ST1	IPE 240	5.65	1	173	R9-R11
ST2	IPE 240	4.10	1	126	R1-R3
B1	L 70x7	6.19	2	91	K1-K2
B2	L 70x7	6.56	2	97	K5-K7
B3	L 70x7	7.81	2	115	K22-K23
B4	L 70x7	5.96	2	88	K25-K26
B5	L 70x7	6.05	2	89	K15-K19
B6	L 70x7	6.10	2	90	K14-K18
B7	L 70x7	7.26	2	107	K20-K27
B9	L 70x7	5.96	2	88	K18-K23
RT-N-W.1	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-N-W.2	M24 rod 8.8	8.22	2	58	panel x 72.1-77.8
RT-N-E.1	M24 rod 8.8	8.37	2	59	panel x 81.8-87.2
RT-N-E.2	M24 rod 8.8	8.38	2	59	panel x 87.2-92.5
RT-N-E.3	M24 rod 8.8	7.10	2	50	panel x 92.5-95.5
RT-JOG.1	M24 rod 8.8	6.30	2	45	panel x 77.8-81.8
RT-W.1	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-W.2	M24 rod 8.8	6.17	2	44	panel x 68.0-72.1
RT-W.3	M24 rod 8.8	5.05	2	36	panel x 68.0-72.1
RT-W.4	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-E.1	M24 rod 8.8	7.16	2	51	panel x 89.9-95.5
RT-E.2	M24 rod 8.8	7.42	2	53	panel x 89.9-95.5
RT-E.3	M24 rod 8.8	8.57	2	61	panel x 89.9-95.5

COLD-FORMED AND TOTALS BY SECTION

Item	Section	Qty	Total m	kg
Hot-rolled / bracing members above	IPE 300	18	82.7	3489
Hot-rolled / bracing members above	IPE 330	1	9.8	481
Hot-rolled / bracing members above	IPE 240	34	204.3	6272
Hot-rolled / bracing members above	HEA 140	28	99.9	2468
Hot-rolled / bracing members above	L 70x7	16	103.8	766
Hot-rolled / bracing members above	M24 rod 8.8	26	188.6	669
Purlins Z200x2.0 @ 1.5 m (110 pcs) + anti-sag	Z200x2.0	110	290.2	1798
Girts Z200x2.0 (58 pcs) incl. sleeves	Z200x2.0	58	281.2	1792
Eave rails C200 (G-W 10.0 + G-E 13.9) + well header 2 x 3.9	C200x60x2.5	4	31.7	219
Sill rail C100x50x3 (D11)	C100x50x3	-	101	404
Upstand framing C100x50x3 (both wells)	C100x50x3	-	90	360

TOTALS

Hot-rolled members 12.7 t + plates 2.3 t = 15.1 t
Wall bracing + roof rods 1.44 t; cold-formed Z/C 4.6 t
STEEL WEIGHT 21.1 t (43 kg/m2 of 486 m2 footprint, 48 kg/m2 of 439 m2 roofed)
Design report Rev 6a s.10: BOM basis 22.1 t (P13 IPE 330 + 70 kg).
Galvanising: hot-dip all members and plates; Z/C S350GD Z275. Bases per bases_C.md Rev 8 (type E, P at K21).

PLATES - estimate

Plates and fittings (S275 unless noted)	Qty	kg
Fin plates FP1 100x150x10 (D1)	53	62.4
Fin plates FP2 160x150x10 (D1, chord splices)	16	30.1
Cap plates 200x280x20 (D2)	27	237.4
Tie plates 140x400x10 (D3, rows F and B)	8	35.2
Base plates 300x400x20 (type E x13 + K21)	14	263.8
Key-pair plates 800x400x20 (K1, K2, K5, K10, K22)	5	251.2
Key-pair plates 400x800x20 (K15, K18, K19, K20)	4	201.0
Key-pair plates K7 400x950, K23 1000x400, K25 / K27 400x850 x20	4	226.1
Key A SHS 90x90x8 x 205 S355 (14 + 13 pairs)	40	166.1
Key B dia 60 x 205 S355 (K3, K4 x2, K6 x2, K8, K14) + WP1	8	36.4
K21 saddle plates 400x150x15	2	14.1
Bracing gussets 10 mm ~350x420 (D4)	32	221.6
Rod gussets 8 mm ~250x250 (D5, 13 panels)	52	122.5
Purlin cleats 120x160x8 (D8)	220	265.3
Girt cleats L100x100x8 x150 (D9)	116	204.9
WP1 base plate 250x250x15 + head cleat	1	9.2

BOLTS AND ANCHORS - estimate

Bolts, bars and anchors (add 5 % spare)	Qty
M20 8.8 x 60 (fin plates FP1 / FP2)	168
M20 8.8 x 70 (cap + tie plates)	108
M20 8.8 x 60 (bracing gussets, 2 per angle end)	64
M16 8.8 (rod gussets 2 per rafter leg; crossing clips)	112
M24 8.8 rods with turnbuckle, nuts + lock nuts	26
M12 8.8 (purlin / girt cleats, fly braces, eave rail)	872
dia 16 B500 post-installed bars M16 threaded, EAD 330087 injection (27 bases x 4)	108
M12 anchors (WP1 location)	2
M8 resin anchors sill rail @ 600 (D11)	170
Key A pockets 140 x 200 cored; Key B pockets 110 x 200	48
Pilot holes 10 x 600 + production holes 20 x 600 (bars)	108

PROJECT	SHEET TITLE	REVISIONS	CURRENT REVISION	DRAWING No.
Steel roof - Administration Building, Regatta Tourist Village, Tripoli	BILL OF MATERIALS	32026-09-29Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a	Rev 6a superstructure Rev 8 bases / bracing Rev 5a	RTV-ST-S06
ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS	Lengths in m from members_C.csv Rev 6a; weights nominal (kg/m x length), plates by volume x 7850	22026-09-28Design Rev 3 / bases Rev 4b (critique C1-C15)	DATE 2026-09-29	S06
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m	Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9	12026-09-27First issue: design Rev 2, bases Rev 3	Drawn: Designed: Checked: Approved:	